

SUMMARY

Data Engineer with 4+ years of experience in Python, SQL, and Bash, specializing in cloud-native ETL/ELT pipelines with Databricks, Kafka, Delta Lake, dbt, and Snowflake to build scalable, compliance-grade systems supporting equity-focused, trauma-informed work in mission-driven organizations. Committed to FERPA and HIPAA-aligned governance that strengthens community trust. Physicist and philosopher by training, I bring an ethics-informed lens to every system I build.

SKILLS

Programming Languages: Python, SQL, Bash, R, JavaScript/TypeScript, YAML, Jinja2, VBA, MATLAB

Analysis & Visualization: Tableau, Power BI, Excel, ArcGIS, Streamlit

Machine Learning: Scikit-learn, PyTorch, LangChain, OpenAI API, Vector Databases (Chroma), RAG, Prompt Engineering

Data Engineering: PySpark, Databricks, Delta Lake, dbt, Kafka, Snowflake, PostgreSQL, MongoDB, Hadoop, Apache Airflow

Cloud & DevOps: GCP (GCS, Dataproc, IAM), AWS (S3, EMR), Azure, Docker, Terraform, GitHub, GitHub Actions CI/CD, Bitbucket

PROFESSIONAL EXPERIENCE

DATA ANALYTICS & ENGINEERING INTERN

Partnership With Children

Jan 2025 - Jan 2026

- Engineered a **FERPA**-compliant automated ETL pipeline in **Python**, **JavaScript**, and **Bash** to migrate protected student records from Apricot into SharePoint, reducing a month-long manual workflow to 7 minutes while maintaining data integrity and real-time access
- Developed a secure **Azure**-hosted **Streamlit** dashboard with **Docker** and **Microsoft SSO** that provided leadership with reliable visibility into program performance, informing grant renewals, staffing allocation, and program expansion decisions across NYC schools, supported by **VBA** automation and **GitHub Actions CI/CD**
- Consolidated 10+ years of NYSED, InfoHub, and internal program datasets using advanced **SQL** joins and unions, validating and reconciling discrepancies to within 0.5% margin of error and standardizing metrics that directly supported grant reporting and district evaluations measuring social work impact across 30,000+ students in underserved NYC communities
- Migrated the entire Apricot case-management platform, including 10+ years of **FERPA** and **HIPAA**-sensitive student records, into Salesforce across 32 NYC schools, partnered with non-technical stakeholders to design **SOQL**-driven notification Flows that fit their actual workflows, and established data governance standards adopted organization-wide

NEUROAI DEEP LEARNING RESEARCHER

Neuromatch

May 2024 - Sep 2026

- Built research data pipelines across three Neuromatch cohorts processing 1.5M+ COCO-MS image captions via image ID joins and one-hot encoding to feed **RoBERTa** and **BERT** models (2024), structured large-scale neural recording data for **GLM**-based behavioral decoding (2025), and implemented a dual forward-pass **Node Perturbation** pipeline benchmarked against **Backprop** and **Hebbian** baselines on **MNIST** (2026), supporting peer-reviewed publication ([arXiv:2409.09560](https://arxiv.org/abs/2409.09560))

DATA ENGINEERING APPRENTICE

Annalect

Aug 2022 - Nov 2022

- Optimized large-scale **Airflow**-orchestrated pipelines for pharma and media clients by restructuring **Jinja2** templates, modifying existing DAGs, and refactoring event-driven **AWS Lambda** trigger logic, reducing data latency by 30%
- Contributed to ETL/ELT workflow automation across AWS (Lambda, **EMR**, **S3**, **Redshift**, **Glue**) and **PostgreSQL**, leveraging **PySpark** and **Redshift Spectrum** to process terabyte-scale data, supporting pipelines driving a 10% CPA drop and 12% performance lift across client pipelines

DATA SCIENCE FELLOW

The Knowledge House

Sep 2021 - Sep 2022

- Built ETL pipelines in Python and Bash to ingest and clean 500+ alumni records, integrated **Google API** for user authentication and data ingestion, and conducted exploratory data analysis, including geospatial engagement patterns using Pandas, supporting a 20% increase in platform user engagement in a cross-functional team environment

PROJECTS

Real-Time Equity Analytics Platform | Kafka, Databricks, Snowflake, dbt

Oct 2025 - Dec 2025

- Designed and deployed a production-grade batch and streaming ELT medallion pipeline ingesting NYC School Climate and CDC Social Vulnerability Index data via Kafka and Dataproc into GCS Bronze/Silver layers, with Databricks Spark Structured Streaming and dbt-modeled Snowflake Gold marts powering district equity dashboards for policy and resource decisions
- Built a LangChain and Chroma RAG service with a FastAPI backend and React/TypeScript UI, embedding 5K+ SVI and school climate documents to enable natural language querying of equity metrics over validated Snowflake Gold data
- Provisioned the entire cloud infrastructure from scratch using Terraform across GCP, Snowflake, and Databricks, implementing DevSecOps practices including detect-secrets scanning, least-privilege IAM roles, and segregated GitHub Actions CI/CD with zero credentials in version control

Automated Compliance-Grade Document ETL Pipeline | Python, JavaScript, Bash

Mar 2025 - May 2025

- Built a FERPA and HIPAA-compliant ETL pipeline in Python, JavaScript, and Bash to extract, validate, and migrate 10,000+ protected student records from Apricot into Excel and SharePoint using Puppeteer-clustered PDF automation and a Tkinter GUI, reducing manual processing from months to minutes

PUBLICATIONS

Krotov, A., Tebo, A., Picart, D., Algave, A. (2024). Evaluating authenticity and quality of image captions via sentiment and semantic analyses. [arXiv:2409.09560](https://arxiv.org/abs/2409.09560)

Picart, D. (2018). Method of Gas Detection Insensitive to Vibration. [Wright's Research Lab](https://www.wrightresearchlab.com/)

EDUCATION

ADELPHI UNIVERSITY

Bachelor of Science, Physics; Minor, Philosophy

2014 - 2018

- Specialization in Engineering, AI, and Data Analysis with advanced mathematics foundations
- Year-Round NCAA D-II Cross Country and Track & Field Athlete

Engineering Fellowships & Continuing Experience

2022 - Present

- Program Earth Software Engineering Fellow (Sep 2024 - Present) · TKH Data Science AEP Fellow, Data Engineering Track (Sep 2024 - Dec 2024) · TEC Innovate Google Fellow (Oct 2024) · CUNY GIS Research (Oct 2023 - Dec 2023)